



IFEJM: New Intuitionistic Fuzzy Expert Judgment Method for Effort Estimation in Agile Software Development

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Abstract

Many agile projects use expert judgment-based methods for estimating effort. Commonly, the judgments made during estimating project features are consensual. However, this will hardly be achieved when a conflict arises between estimators. Besides, estimate depends on the experience and skills of the estimator and could be threatened by his uncertainty to make reliable and accurate assessments. To fill these gaps, an intuitionistic fuzzy expert judgment method is proposed. The latter allows making fuzzified assessments and integrates estimators' priorities according to a set of human factors. As well, it provides consensual estimates either by the end of the estimation rounds or automatically using an iterative algorithm. On the other hand, an initial empirical study has been conducted on an agile project in which user stories have been estimated by students and experts. The first findings have revealed that the proposal is more suitable for inexperienced estimators or in the first sprints of the project where disagreement is still significant. Nevertheless, when group agreement is increased during the estimating process, the proposal maintains a null bias toward the overestimation or the underestimation of the user stories.

Keywords Effort estimation · Intuitionistic fuzzy · User stories · Expert judgment · Agile projects · Consensus

1 Introduction

Effort estimation is a critical activity required for either conventional or Agile Software Development (ASD) projects. Indeed, sprint and release plans are made based on the estimated effort. Hence, reliable estimation leads to reliable planning that decreases the risk of cost and time overruns. The reliability of estimation closely depends on the method adopted to assess effort [1]. Commonly, there are two main families of methods to assess effort in software engineering [2]. The first family is the formal estimation models, which use quantitative variables or parameters to estimate the time or effort required to complete tasks, for example, lines of code, the number of graphical interfaces or classes used in an object-oriented project, etc. These parameters are deduced from data analysis of past projects and adjusted to align with the specificities of the current project [3]. Whereas, the second family concerns expert judgment-based methods

whose effort measurement is driven by experts' opinions. It requires consulting the expert who will perform a project task to ask him how much time or effort is needed to complete it. In principle, the task description is discussed between the responsible for its accomplishment until they get a mutual agreement or consensus on the task estimation. This technique relies on our trust in the expert's skills and experience. Many surveys [4–6] revealed that the second family of techniques is the most frequent in industry. A more recent survey [7] showed that expert judgment, planning poker, and analogy are the widely practiced techniques in ASD. The same study confirmed that these techniques align well with agility principles since they privilege people and communication.

Within the ASD context, estimation is about quantifying the amount of work or the effort required to accomplish user stories [8]. A user story is a description of a software feature often written in a specified template from an end-user viewpoint [9]. It constitutes the smallest unit of work in an agile project. User stories often written and prioritized in a particular document, called product backlog, are added to sprints or short iterations (from 2 to 4 weeks). At the end of a sprint, its stories should be completed to get a new software increment. Otherwise, they could be moved to the upcoming sprints.

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The user stories are estimated by the agile team (i.e., the project members in charge of the user stories implementation and estimation). Each team member assigns points to each story indicating the effort required for its implementation. Usually, the Fibonacci sequence is used as a scoring scale for expressing the story points [10], e.g., 0.5, 1, 2, 3, 5, 8, 13, 21, and so on.

The choice of the Fibonacci sequence is motivated by its popularity as the most common scale used for estimating user stories in agile projects [11]. This sequence allows relative sizing of work. Indeed, each story is sized relative to the others in the product backlog. For this reason, a baseline story could be identified to compare all stories against. For instance, a story that is assigned three points should be triple as much as a story that is assigned one point. However, using real numbers in estimation does not tolerate comparison of stories' sizes [10]. Moreover, the Fibonacci values increase gradually the estimates of stories when their complexities or sizes are increased too [12]. This fact enlightens the inherent growth of uncertainty in estimating from small to large stories.

This paper focuses on expert judgment-based methods in the ASD effort estimation realm. Expert judgment is more suitable for companies that lack historical data from previous projects or have restricted estimating resources [13]. Both situations are present in agile projects. Uncertainty and inconsistency are the main problems facing expert judgment [14]. Indeed, there are many software projects with high inherent uncertainty [15]. The uncertainty could arise from uncertain and vague requirements, for which the work could not be perfectly specified. Even worse, when a conflict arises between experts, a failure of the method is expected, or at least a huge amount of time might be wasted to get a consensus. Jørgensen in [16] suggested that this kind of problem should be mechanically integrated within the expert judgment. Another study conducted by Faria and Miranda [17] showed that an unstructured expert judgment method could lead to inconsistent judgments. The authors recommended that these methods should be well described with systematic and detailed steps. In this way, they could benefit from the following advantages:

- **Correctness:** evaluating whether the method respects or not a prescribed process.
- **Traceability:** understanding how the outputs were derived from their inputs.
- **Reduced variability:** reducing variability and dispersion in experts' judgments with a well-defined process.

Furthermore, the expert judgment-based method is a subjective technique that heavily relies on expert intuition [18]. This intuition is usually built from his acquired skills and past experiences in similar projects [7,14]. Additionally, the

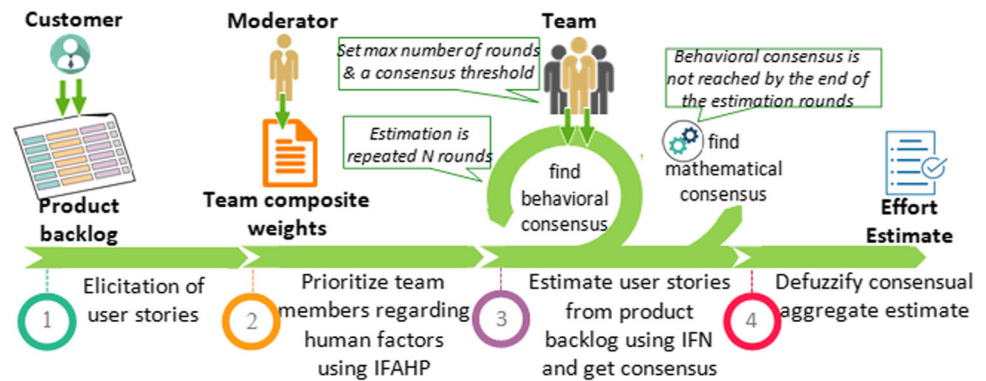
experts who have presented the best accurate assessments in previous sessions should not be put on an equal footing with others. Therefore, experts' priorities should be established with regard to those human factors. In this way, we propose an intuitionistic fuzzy expert judgment method (IFEJM) to estimate effort in the ASD context. The objective of the IFEJM is to provide a structured method of extracting, capturing, and combining expert judgments and their priorities in a fuzzy environment.

The proposed method (see Fig. 1) starts by deriving the experts' priorities and calculates afterward the collective estimate from those made by experts at different rounds. A consensus index is set to check the closeness degree of individual estimations to the collective one. When consensus is not reached by the end of the estimation rounds, an iterative algorithm based on a nonlinear programming model is performed. The algorithm aims to slightly adjust individual and collective estimates to automatically reach the consensus. Finally, the consensual collective estimate is defuzzified into a Fibonacci value. The proposal integrates intuitionistic fuzzy logic to handle uncertainty in experts' estimations. The authors Idri et al. [1] argued that the estimation accuracy could be improved when fuzzy logic is combined with the estimation technique. In our study, we have chosen intuitionistic fuzzy logic instead of fuzzy logic since the first one gives more flexibility in handling uncertainties. Here, in addition to the membership degree, experts could use a non-membership degree and a hesitation degree (also known as an indeterminacy degree) to represent their imperfect knowledge of sizing works.

To establish experts' priorities with regard to the human factors, we propose further an extension of the analytic hierarchy process (AHP) [19]. The extension integrates the intuitionistic fuzzy to AHP and puts forward a novel score function to defuzzify intuitionistic fuzzy numbers (IFNs). The choice of AHP can be justified by the ability and efficiency of this tool for weight derivation [20]. Concerning the consensus problem, it would be more logical and easier for experts to get agreement about setting a consensus threshold, rather than attempting to get consensus when estimating each user story. Although working in ASD should be done in a collaborative and friendly way, the emergence of conflicts in agile teams is very probable and could be harmful to the estimate session. This paper did not deal with team conflict resolution, but it proposes a solution to get consensus in such situations. To that, the proposal has been initially validated on an agile project.

The rest of the paper is organized as follows: Sect. 2 presents the related works. Section 3 introduces the preliminaries of this work. Section 4 outlines the methodology background by discussing the consensus issue and the critical human factors influencing effort estimate in ASD. Section 5 presents the proposed method, while Sect. 6 describes its

Fig. 1 The main steps of the IFEJM for effort estimation in ASD



different steps. In Sect. 7, an illustrative example is given to show the proposal's applicability. In Sect. 8, an experiment is conducted to validate and compare the results of the proposal with those of other studies. Section 9 is devoted to concluding the paper.

2 Related Works

Usman et al. [7] revealed that the expert judgment method is one of the most practiced techniques for effort estimation in ASD. Usually, some lightweight expert judgment techniques are used by agile practitioners, like planning poker, dot voting, and T-Shirt sizes. Despite the simplicity of these techniques, they are still unable to deal with all problems related to effort estimation in ASD, such as uncertainty and conflict. Even so, the most proposed approaches in the literature are formal models or machine learning-based methods. Few research studies have been interested in new structured expert judgment methods that deal with uncertainty and conflicting situations and integrate human factors in estimating the efforts of ASD projects.

Klaus-Rosińska et al. [21] proposed an expert estimation approach based on type-2 fuzzy numbers (T2FN). User stories are evaluated by different teams using “Effort points” and “Priority points.” The consensus is obtained mathematically by the intersection of the fuzzy intervals of estimates. However, the study marginalized the behavioral consensus although its importance in ASD.

Torrecilla-Salinas et al. [22] investigated a new Delphi-based consensus-making method in the web engineering field. They used a set of pre-defined questionnaires to gather experts' judgments expressed on the Likert scale. Afterward, the organizers provide the experts with feedback, including statistical analysis such as the percentage of agreement and disagreement. Based on this feedback, experts review their own judgments during the next rounds. However, this study did not address the situation of persistent disagreement, even after feedback and review.

Miranda et al. [23] proposed an expert estimation method in ASD called full factorial pair-wise comparison. Indeed,

each user story is compared against the others using a verbal scale and recorded in a judgment matrix. Next, based on an inconsistency index and a standard deviation, judgments are revised until consistency of the matrix. Yet neither behavioral nor mathematical consensus was treated in the case of several judgment matrices.

Rola and Kuchta [24] proposed a fuzzy expert estimation method for Scrum projects. They used triangular fuzzy numbers (TFN) for expressing both experts' estimations and weights. An expert weight represents his agreement degree with the team about a user story estimation. These weights should be accepted by all experts and for all user stories. Otherwise, experts' weights will be imposed when no agreement is met after a pre-set period. This manner of reaching a consensus does not respect the principle of self-organized teams in ASD. As well, the study did not support relative estimation.

An expert judgment method proposed by [25] used fuzzy AHP to predict the efforts of software projects, including agile ones, from a reference project. Experts provide pair-wise comparisons between projects using TFNs to derive their weights and deduce thus their efforts. However, the absence of a reference project makes the method applicability difficult, if not impossible. Moreover, the paper did not address the consensus issue, especially in the case of many pair-wise comparison matrices made by experts for a single project.

Compared to our proposal, despite the crucial role of human factors in expert judgment methods, none of the studied approaches integrated them. They also did not provide a limit of rounds to get behavioral consensus or at least could waste a lot of time to reach it, despite timeboxing story estimation such in [24]. Only two approaches [22] and [24], in addition to the proposal, combine behavioral and mathematical consensus. The complete comparison is summarized in Table 1.



Table 1 Comparison of expert judgment-based methods estimating effort in ASD

Approach	Criteria					
	Uncertainty & inconsistency	Human factors	Rating scale	Relative estimate	Consensus type (Beh ¹ / Math ²)	Duration of reaching Beh ¹ consensus
Proposed Approach	Both	Yes	IFN	Yes	Both	Limited by a max no. of rounds and a consensus threshold
Approach of [21]	Both	No	T2FN	Yes	Math	NA
Approach of [22]	Uncertainty	No	Likert	Yes	Both	Unlimited
Approach of [23]	Inconsistency	Yes	Verbal	Yes	NA	NA
Approach of [24]	Both	No	TFN	No	Both	Limited by timeboxed stories
Approach of [25]	Both	No	TFN	Yes	NA	NA

The inconsistency is supported in our approach through IFAHP, which avoids inconsistent preferences of experts and factors

¹Behavioral consensus reached through discussion between experts

²Mathematical consensus reached through computations and rules

3 Preliminaries

3.1 Intuitionistic Fuzzy Sets

This section briefly introduces the intuitionistic fuzzy set (IFS) and gives the main definitions and notations used in this paper. Atanassov extended the fuzzy logic with the notion of IFS since 1986 [26]. Compared to fuzzy sets, the IFS supports intuition in human judgments by proposing a supplementary function called non-membership degree.

Definition 1 [26] Let A be an IFS of a given set X , A is defined as follows:

$$A = \{(x, \mu_A(x), \nu_A(x)) \mid x \in X\} \quad (1)$$

The functions $\mu_A(x)$ and $\nu_A(x)$ return the degrees of membership and non-membership of every $x \in X$ to the set A , respectively. The following conditions should be respected: $0 \leq \mu_A(x) \leq 1$, $0 \leq \nu_A(x) \leq 1$ and $0 \leq \mu_A(x) + \nu_A(x) \leq 1$.

Besides, $\pi_A(x) = 1 - \mu_A(x) - \nu_A(x)$ is called the intuitionistic fuzzy index, known as the degree of indeterminacy or hesitation of x to A . Obviously, we have $0 \leq \pi_A(x) \leq 1$, for all $x \in X$. Every fuzzy set can be expressed as an IFS [27], and it suffices that $\nu_A(x) = 1 - \mu_A(x)$, and $\pi_A(x) = 0$.

Xu [28] proposed to write an IFN as $\tilde{\alpha} = (\mu_{\alpha}, \nu_{\alpha})$ instead of the old notation. As well, Xu proposed some operations performed on IFNs as follows:

Definition 2 [28] Let the two IFNs $\tilde{\alpha}_1 = (\mu_{\alpha_1}, \nu_{\alpha_1})$ and $\tilde{\alpha}_2 = (\mu_{\alpha_2}, \nu_{\alpha_2})$, the following operations are defined:

- $\tilde{\alpha}_1 \cap \tilde{\alpha}_2 = (\min(\mu_{\alpha_1}, \mu_{\alpha_2}), \max(\nu_{\alpha_1}, \nu_{\alpha_2}))$
- $\tilde{\alpha}_1 \cup \tilde{\alpha}_2 = (\max(\mu_{\alpha_1}, \mu_{\alpha_2}), \min(\nu_{\alpha_1}, \nu_{\alpha_2}))$
- $\tilde{\alpha}_1 \oplus \tilde{\alpha}_2 = (\mu_{\alpha_1} + \mu_{\alpha_2} - \mu_{\alpha_1} \times \mu_{\alpha_2}, \nu_{\alpha_1} \times \nu_{\alpha_2})$

$$4. \tilde{\alpha}_1 \otimes \tilde{\alpha}_2 = (\mu_{\alpha_1} \times \mu_{\alpha_2}, \nu_{\alpha_1} + \nu_{\alpha_2} - \nu_{\alpha_1} \times \nu_{\alpha_2})$$

$$5. \lambda \tilde{\alpha}_1 = (1 - (1 - \mu_{\alpha_1})^\lambda, \nu_{\alpha_1}^\lambda), \lambda \geq 0$$

Various measures are available to assess the similarity distance between two IFNs as advanced by Hung and Yang [29]. Based on the Hamming and the Euclidean distances, Atanassov [30], Szmidt and Kacprzyk [31] proposed several methods to calculate the distance between two IFNs. Motivated by the Hamming distance, this distance is calculated as follows:

$$d(\tilde{\alpha}_1, \tilde{\alpha}_2) = \frac{1}{2}(|\mu_{\alpha_1} - \mu_{\alpha_2}| + |\nu_{\alpha_1} - \nu_{\alpha_2}| + |\pi_{\alpha_1} - \pi_{\alpha_2}|) \quad (2)$$

Definition 3 [32] An intuitionistic fuzzy preference relation (IFPR) is noted $R = (\tilde{r}_{ij})_{n \times n}$ where $\tilde{r}_{ij} = (\mu_{ij}, \nu_{ij})$ are IFNs. μ_{ij} indicates the degree to which an element x_i is preferred to x_j , while ν_{ij} indicates the degree to which the element x_i is not preferred to x_j . As well, μ_{ij} and ν_{ij} satisfy the conditions:

$$\mu_{ij} = \nu_{ji}, \quad \mu_{ji} = \nu_{ij}, \quad \mu_{ii} = \nu_{ii} = 0.5 \\ \mu_{ij}, \nu_{ij} \text{ and } \mu_{ij} + \nu_{ij} \in [0, 1], \forall i, j = 1, \dots, n \quad (3)$$

The IFPRs should be consistent before deriving priorities, or else judgments are considered unreliable. Two types of consistency have been proposed for IFPRs in the literature: additive and multiplicative. However, the multiplicative consistency is more suitable for IFPRs than the additive consistency, because of the deficiency of the additive consistency with the ratio scale 0.1–0.9 [33].

Definition 4 [33] An IFPR $R = (\tilde{r}_{ij})_{n \times n}$ is called multiplicative consistent if the following multiplicative transitivity is



verified:

$$\mu_{ij}\mu_{jk}\mu_{ki} = v_{ij}v_{jk}v_{ki} \text{ for all } i, j, k = 1, \dots, n \quad (4)$$

In real situations, it is very complex for experts to create IFPRs satisfying the previous multiplicative transitivity. Hence, the experts can use a consistency index to check the consistency of their IFPRs. In this paper, we use the multiplicative consistency index proposed by Wan et al. [34]:

$$CI(R) = \frac{1}{n(n-1)(n-2)} \sum_{1 \leq i < j < k} \left| \ln \left(\frac{v_{ij}}{\mu_{ij}} \right) - \ln \left(\frac{v_{ik}}{\mu_{ik}} \right) - \ln \left(\frac{v_{kj}}{\mu_{kj}} \right) \right| \quad (5)$$

$\frac{v_{ij}}{\mu_{ij}}$ can be interpreted as the ratio of the degree of alternative x_i non-preferred to x_j to that of alternative x_i preferred to x_j , and so on for the other ratios. CI is calculated based on the deviation between the natural logarithms of these ratios. In particular, when $CI(R) = 0$, the IFPR R is considered completely multiplicative consistent. Similar to the Saaty method [19], we choose 0.1 as a consistency threshold. In this way, if $CI(R) \leq 0.1$, then R is considered of acceptable multiplicative consistency. The preferences in the IFPRs have to be corrected until the condition is satisfied. Sometimes owing to the lack of time, IFPRs could be automatically adjusted to reach consistency [35,36].

3.2 Intuitionistic Fuzzy AHP

AHP presented by Saaty [19] is a multi-criteria decision-making (MCDM) method that structures the problem into a multilevel hierarchy of goals, criteria, sub-criteria, and alternatives. AHP is widely used in different domains and is well known as a straightforward MCDM approach that can decrease the complexity of decision-making problems by decomposition, as well as an efficient tool for weight derivation [20]. Another motivation is related to the AHP's ability to support pair-wise comparisons for making numerical assessments of qualitative phenomena [19], which is the case here. An additional advantage of the method is using relative evaluations, which aligns with the general principles and guidelines of relative estimation in ASD.

However, despite its reputation, AHP does not support dealing with vagueness and uncertainty in decision-making problems. Fuzzy logic could be used with AHP, but the membership function remains insufficient to manage intuition in experts' judgments. A promising solution could be the intuitionistic fuzzy logic which provides in addition to the membership function, the non-membership, and hesitancy functions. For this reason, we propose an extension of the

conventional AHP to the intuitionistic fuzzy AHP (IFAHP). In the literature, many works interested in IFAHP [37–39] were proposed for different purposes, while in this paper it will be used to derive experts' weights.

4 Methodology Background

4.1 Consensus Theory and Its Application for Estimation in ASD

Consensus is not a new concept or notion. It has been examined for many decades in different domains and contexts. In sociology, the most common definition adopted by sociologists was "Consensus is the adjustment of social dissension" [40]. Initially, the consensus was defined as the unanimous agreement of experts on a subject [41]. However, in some cases, consensus fulfillment could be too costly, if not impossible [42]. The authors in [24] considered the consensus as an agreement favorable for the whole group. In other words, a consensual decision is not necessarily a unanimous decision but rather an acceptable decision by experts. In the literature, there are two main approaches to reaching consensus [43]: behavioral and mathematical.

Within a behavioral approach, the experts are brought together at the same place to discuss and come to a consensus. Meanwhile, in a mathematical approach, mathematical rules are used to combine experts' judgments. The latter approach is convenient in conflict situations. In agile projects, many behavioral consensus-based methods have been used, such as Delphi [44], Wideband Delphi [45], and Planning poker [46]. Furthermore, different quick scenarios for solving the consensus problem have been proposed while applying these methods.

For instance, in [10], the author claimed that the agile team should postpone stories that cause a conflict beyond the current estimate session until further information is available. This proposition, in our opinion, is only a problem postponement. Grenning [46] suggested that we could choose the lowest estimate when disagreement persists. However, this suggestion leads to an underestimation of effort and cost, which is considered in the theory of social judgment the worst compared to overestimation [47]. As a common rule applied by many agile teams, the highest estimate [48] is selected. Likewise, this scenario is not the best solution as we will demonstrate in our experiment. In light of the problems related to group estimation, the average could be another alternative [49]. However, this could hide some valuable details related to the implementation of stories [50]. On the other hand, weighted aggregation and Bayesian aggregation are the main techniques used in mathematical consensus-based methods [43]. Notice that all these scenarios of solving the consensus problem are still crisp and do not involve any fuzzy specification.



In fuzzy contexts, and to the best of our knowledge, few research studies have been interested in the consensus problem for effort estimation in ASD. Independently of agile projects, the initial studies conducted by Szmidt and Kacprzyk [31,51,52] introduced different fuzzy similarity measures to assess consensus. Consensual judgment in [53] is obtained through a geometric aggregation of expert opinions represented by interval-valued trapezoidal fuzzy numbers. More recently, the authors of [54] presented a consensus index to check whether the opinions of two experts are consistent or not. Experts' opinions are expressed with interval-valued IFNs. Many of the discussed fuzzy propositions for reaching consensus could be adopted for agile projects. However, a special emphasis should be placed on the agility practices, such as communication, timeboxed estimate sessions, volatile requirements, and relative estimates. Sections 6.5 and 6.6 describe our approach that combines behavioral and mathematical consensus in a fuzzy context.

4.2 Identification of the Human Factors Influencing Estimation in Agile Projects

Commonly in software engineering, the reasons for inaccurate effort estimates could be of technical aspect or human aspect. The technical aspect relates to the estimation models or methods, while the human aspect is more centered on the estimator. In expert judgment methods, the human aspect should be considered more interesting than the technical aspect since the estimate is based on the expert's intuition and subjectivity. Different systematic literature reviews have been conducted on the human factors facing effort estimation in ASD. In [55], the authors claimed that skills and prior experience are the two main human cost drivers or factors. In another survey, Usman et al. [7] focused on the state of the practice of effort estimation in agile teams (60 agile practitioners issued from 16 countries). The study showed that practitioners consider skills (65%) and experience (72%) the most common factors utilized for effort estimation in ASD. Skitmore and Lowe [18] also confirmed that experience is a prominent factor for accurate estimates. However, according to the same authors, the estimator's experience should not be restricted to his current participation in an activity. It must besides comprise learning from the group and correcting errors in judgments. For this reason, integrating the individual estimate accuracy in the proposed approach as a human factor is crucial to evaluate the learning rate of each expert from the last estimate sessions. Stupar et al. [56] proposed a fuzzy expert system to estimate effort in the Scrum method, including accuracy or quality of estimation. The latter is recalculated after each sprint to give a more accurate prediction over time.

Even in formal models considered more objective, different human factors that affect effort estimation have been

proposed. For instance, in the COCOMO II model [57], the human factors are personnel continuity, analyst, and programmer capabilities, experience with application, language, and tool experiences. In the use cases points method [58], some human factors were included as environmental cost drivers, such as lead analyst capability, the motivation level of the team, and their experience with object-oriented applications. Among the human factors previously proposed in the literature, we might distinguish the main dominant ones: experience, analysis skills, programming skills, and individual accuracy of estimation.

The experience herein is related to the years spent by an expert working on similar agile projects. The analysis skills concern writing user stories, scenario analysis, modeling, and designing patterns and architecture. The programming skills comprise code implementing, testing, integrating, deploying, and database engineering. Individual accuracy of estimation is the degree to which the effort estimated by an expert for a set of user stories varies from the actual one deployed to their accomplishment in a completed sprint.

5 Proposed Method

The proposal presents a new structured expert estimation-based method that deals with the estimate process in uncertain and conflicting contexts (see Fig. 2). It begins with an initial step whereby a moderator of the estimate session makes preferences between human factors and between estimators according to those factors. The objective is to derive global or composite estimators' weights. During this preliminary step, the agile team should also set a consensus threshold and a maximum number of rounds to begin estimation. The initial values of these parameters could be either kept throughout the estimate sessions or modified upon the arrival of new stories and the team deems it necessary.

The first round starts at the moderator's signal, and the estimators estimate a collection of user stories (should not exceed 7 ± 2 user stories) chosen arbitrarily from the product backlog. This limit was approved in psychology by Miller [59]. The author affirmed that an ordinary human could simultaneously hold up this number of objects. The estimators rate stories, introduced by the moderator one by one, using linguistic terms equivalent to IFNs that will be presented later in Sect. 6.3. At this step, the estimators could ask questions and request more details about user stories. A round of estimation can also be timeboxed to avoid spending a lot of time. The estimators continue to rate user stories without revealing their estimates. The objective is to create an individual effort estimation vector for this collection of user stories (estimation per lot). The estimators could further choose in agreement a baseline user story among those that are introduced and modify their estimates as long as the round is not finished.



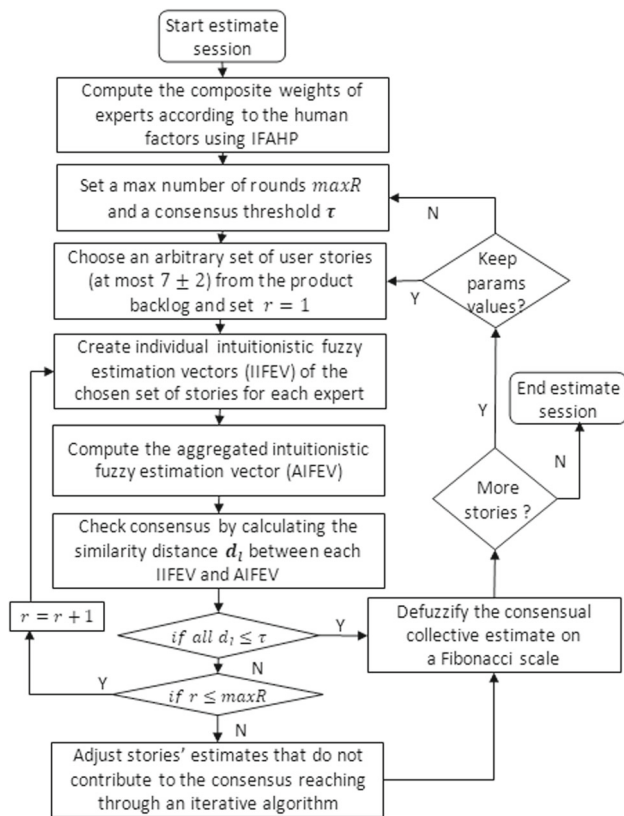


Fig. 2 Flowchart of the IFEJM

Notably, the estimation per lot will improve the velocity and analogy in the estimating process.

At the end of each round, the individual estimates are aggregated based on the global estimators' weights to get the collective estimate, and the consensus is checked afterward. Indeed, as soon as all individual estimates are close to the collective one, i.e., at a distance less than the consensus threshold τ , the collective estimate is acceptable and assigned to the set of selected user stories. Otherwise, the moderator triggers another round as long as the maximum number of rounds is not attained. The estimators, at this new round, are invited to discuss and revise their estimates and try to move toward a consensus. In particular, they should discuss the stories not contributing to its achievement (i.e., the stories having distances from the aggregate estimate superior to the consensus threshold).

When all estimate rounds are completed without meeting the consensus, an iterative algorithm is performed to adjust slightly collective and individual estimates of the last round. Finally, the consensual collective estimate got either directly or automatically is defuzzified into Fibonacci values. The estimators can repeat this process until they estimate all user stories in the product backlog.

In this context, the effort estimation problem in ASD can be described as: $U = \{u_1, u_2, \dots, u_p\}$ is the set of user

stories estimated by experts $D = \{d_1, d_2, \dots, d_m\}$, with different priority weights $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_m\} (\lambda_i \in [0, 1])$. The weights are derived from pair-wise comparisons between experts with respect to a finite set of human factors $F = \{f_1, f_2, \dots, f_n\}$. Stories are estimated at different rounds $R = \{r_1, r_2, \dots, r_q\}$, where q does not exceed an upper limit of rounds $maxR$. The objective is reaching a consensual collective estimate, i.e., at a similarity distance to the individual estimates less than a consensus threshold τ .

6 Description of the Proposal Steps

6.1 Computing Global Experts' Weights Using IFAHP

The aim is to privilege the experts or estimators with more experience, expertise, and accuracy in prior estimates. Hence, the proposal starts by deriving human factors priorities. Next, in the same way, the local experts' priorities according to each factor are determined. Finally, global or composite weights of experts are calculated using a novel score function to defuzzify IFNs.

6.1.1 Derivation of Human Factors Priorities

The moderator of the estimate session should compare the human factors to each other to provide the corresponding pair-wise comparison matrix. For this reason, the 0.1–0.9 ratio scale in Table 2 is used. Using Eq. (5), the multiplicative consistency of the factors comparison matrix should be checked against the consistency threshold set to 0.1 as [19].

Table 2 Scale for rating the importance of human factors and experts in agile projects [60]

0.1-0.9 scale	Description
0.1	Extremely not preferred
0.2	Very strongly not preferred
0.3	Strongly not preferred
0.4	Moderately not preferred
0.5	Equally preferred
0.6	Moderately preferred
0.7	Strongly preferred
0.8	Very strongly preferred
0.9	Extremely preferred
Other values between 0 and 1	Intermediate values of the above numbers



The pair-wise comparison matrix $Mf = (\tilde{f}_{ij})_{n \times n}$ is an IFPR of factors expressed as follows:

$$Mf = \begin{matrix} & f_1 & \cdots & f_n \\ \begin{matrix} f_1 \\ \vdots \\ f_n \end{matrix} & \begin{pmatrix} (\mu_{11}, v_{11}) & \cdots & (\mu_{1n}, v_{1n}) \\ \vdots & \ddots & \vdots \\ (\mu_{n1}, v_{n1}) & \cdots & (\mu_{nn}, v_{nn}) \end{pmatrix} \end{matrix}$$

$\tilde{f}_{ij} = (\mu_{ij}, v_{ij})$ is an IFN, where μ_{ij} represents the degree to which the factor f_i is preferred to the factor f_j , while v_{ij} represents the degree to which the factor f_i is not preferred to the factor f_j . Once Mf is multiplicative consistent, the priority weights of human factors could be calculated. Indeed, let $\tilde{w}_{f_k} = (\mu_{f_k}, v_{f_k})$ be the intuitionistic fuzzy weight of the k th factor. To calculate μ_{f_k} and v_{f_k} , the following equations are used:

$$\mu_{f_k} = \frac{1}{n} \sum_{j=1}^n \left(\frac{\mu_{kj}}{\sum_{i=1}^n \mu_{ij}} \right) \quad (6)$$

$$v_{f_k} = \frac{1}{n} \sum_{j=1}^n \left(\frac{v_{kj}}{\sum_{i=1}^n v_{ij}} \right) \quad (7)$$

6.1.2 Derivation of Experts Local Priorities

The rating of experts' preferences needs to be transparently and rationally done by the moderator. In fact, for each human factor, experts are compared to each other. Consequently, the pair-wise comparison matrix $Md^{(k)} = (\tilde{d}_{ij}^{(k)})_{m \times m}$ is created.

$$Md^{(k)} = \begin{matrix} & d_1 & \cdots & d_n \\ \begin{matrix} d_1 \\ \vdots \\ d_n \end{matrix} & \begin{pmatrix} (\mu_{11}^{(k)}, v_{11}^{(k)}) & \cdots & (\mu_{1n}^{(k)}, v_{1n}^{(k)}) \\ \vdots & \ddots & \vdots \\ (\mu_{n1}^{(k)}, v_{n1}^{(k)}) & \cdots & (\mu_{nn}^{(k)}, v_{nn}^{(k)}) \end{pmatrix} \end{matrix}$$

where m is the number of experts. $Md^{(k)}$ is an IFPR of experts according to the k th factor f_k . $\tilde{d}_{ij}^{(k)} = (\mu_{ij}^{(k)}, v_{ij}^{(k)})$ is an IFN representing experts' preferences. The IFPR is performed using the same scale of Table 2, and its multiplicative consistency is checked in the same manner as the factor pair-wise comparison matrix. The local priority weight of an expert according to the factor f_k is an IFN denoted as $\tilde{w}_{d_i, f_k} = (\mu_{d_i, f_k}, v_{d_i, f_k})$. Its membership and non-membership are also calculated using Eq. (6) and Eq. (7).

6.1.3 Global Weights of Experts

Crisp composite or global weights of experts should be determined to calculate their aggregate estimation. As such, based on the local expert weight regarding a factor and the corresponding factor weight, the intuitionistic fuzzy composite

weight of an expert is calculated using Eq. (8):

$$\widetilde{GW}_l = \tilde{w}_{d_l, f_1} \otimes \tilde{w}_{f_1} \oplus \tilde{w}_{d_l, f_2} \otimes \tilde{w}_{f_2} \cdots \oplus \tilde{w}_{d_l, f_n} \otimes \tilde{w}_{f_n} \quad (8)$$

Afterward, the intuitionistic fuzzy composite weight should be defuzzified to get the crisp global weight of an expert. For this reason, we propose new score functions S , S^+ and $\neg S^-$ deduced from a voting matrix (Table 3).

Inspired by voting problems, the main idea of this matrix is to propose a simple tool that allows defuzzifying IFN. The matrix values are fed from degrees of membership and non-membership of the IFN, reflecting thus the percentages of voters for and against. The defuzzified number is calculated from the proposed matrix using score functions representative of voting results.

In other words, an IFN $\tilde{\alpha} = (\mu, v)$ could be interpreted as follows: The membership degree μ is the percentage of persons voted for, the non-membership v is the percentage of persons voted against, and the indeterminacy degree π is the percentage of neutrality or absenteeism. For example, in an electoral system, with 10 voters, the IFN (0.5, 0.3) means that 5 persons voted for and 3 voted against, while 2 persons were absent. The binary label "Option" in the voting matrix distinguishes the two options of votes ("For" and "Against"). Meanwhile, the "Vote" column indicates positive and negative percentages of voters according to these options.

On this basis, the positive score S^+ is the ratio of the voters who have voted for to the sum of all voters. The non-negative score $\neg S^-$ is the ratio of voters who have not voted against to the sum of persons who have voted neither for nor against. Accordingly, S^+ and $\neg S^-$ of the IFN are expressed as follows:

$$S^+(\tilde{\alpha}) = \frac{\mu}{\mu + v}, \text{ with } \mu + v > 0 \quad (9)$$

$$\neg S^-(\tilde{\alpha}) = \frac{\mu + \pi}{\mu + \pi + v + \pi} = \frac{\mu + \pi}{1 + \pi} \quad (10)$$

subject to: $S^+(\tilde{\alpha})$ and $\neg S^-(\tilde{\alpha}) \in [0, 1]$.

The final score function $S(\tilde{\alpha})$ is the mean of the previous scores. Among several combination operators, the arithmetic and the geometric means are often used to combine the membership and non-membership of IFNs with a single value [37].

In this paper, the geometric mean is chosen as the combination operator to obtain the score function $S(\tilde{\alpha})$ as follows:

$$S(\tilde{\alpha}) = \sqrt{S^+(\tilde{\alpha}) \neg S^-(\tilde{\alpha})} = \sqrt{\frac{\mu(\mu + \pi)}{(\mu + v)(1 + \pi)}} \quad (11)$$

If $S(\tilde{\alpha}_1) < S(\tilde{\alpha}_2)$, then $\tilde{\alpha}_1$ is smaller than $\tilde{\alpha}_2$ denoted as $\tilde{\alpha}_1 < \tilde{\alpha}_2$.



Table 3 Voting matrix of an IFN

Option	Voting	
	Positive	Negative
For	μ	$\nu + \pi$
Against	ν	$\mu + \pi$

If $S(\tilde{\alpha}_1) = S(\tilde{\alpha}_2)$, then $\tilde{\alpha}_1$ is compared to $\tilde{\alpha}_2$ using the accuracy function of [28]: $H(\tilde{\alpha}) = \mu + \nu$.

Based on the score function of Eq. (11), the intuitionistic fuzzy global weights are normalized to get the crisp vector of global experts weights: $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_m)^T$

$$\lambda_l = \frac{\sqrt{\frac{\mu_l(\mu_l + \pi_l)}{(\mu_l + \nu_l)(1 + \pi_l)}}}{\sum_{l=1}^m \sqrt{\frac{\mu_l(\mu_l + \pi_l)}{(\mu_l + \nu_l)(1 + \pi_l)}}} \quad (12)$$

subject to: $\lambda_l \in [0, 1]$ and $\sum_{l=1}^m \lambda_l = 1$.

6.2 Setting of the Consensus Threshold and the Max Number of Rounds

Before estimating stories, the agile team should set a max number of rounds $maxR$ and a consensus threshold τ . Unless otherwise decided, the initial values of these parameters could be kept throughout the estimate session. They could also be modified many times during the same session. The modification might be motivated by either the need to accelerate or decelerate the team's velocity. Besides, some guidelines are proposed to set adequate $maxR$ and τ .

It is common to hold the estimate session two times: at the start of the project and during the project once by sprint. At the start of the project, when the product backlog is ready, the value of $maxR$ should be chosen smaller, and the value of τ greater than those of the estimate session held once by sprint. In fact, at the start of the project, the estimate is rough and specifically used to prioritize the product backlog and provide the release plan. However, when new items are added, deleted or even changed during sprints ("grooming operation" [61]), the agile team should give itself more chances to discuss changes. Contrary to the start of the project, during sprints, the value of $maxR$ should be chosen greater and τ smaller than those used when the product backlog is initially written.

It is important to note that as long as the $maxR$ value is great, communication will be enhanced between experts and chances to get behavioral consensus are high. By doing this, we respect one of the most relevant principles of the agility "face-to-face conversation" claimed by the agile manifesto [62]. Anyway, choosing a small value of τ avoids quickly stopping discussion between experts from early rounds and

Table 4 Linguistic terms and their corresponding IFNs for rating user stories efforts

Linguistic terms	IFN	Fibonacci scale
Extremely low effort (ELE)	(0.10, 0.90)	1
Very low effort (VLE)	(0.15, 0.80)	2
Low effort (LE)	(0.25, 0.65)	3
Medium effort (ME)	(0.45, 0.40)	5
High effort (HE)	(0.70, 0.10)	8

being satisfied with aggregating their estimates, even if they are strongly divergent.

6.3 Creation of Individual Intuitionistic Fuzzy Estimation Vector

As long as the max number of rounds and the consensus are not reached, the experts are invited to estimate the set of user stories using linguistic terms as shown in Table 4. Notice that the proposed scale gradually increases the hesitation degree, reflecting thus the high uncertainty of the expert to estimate large and complex stories. Indeed, we assign an IFN with a low hesitation degree to the user story because we are more confident of its context, difficulties, and properties. Otherwise, an IFN with a high hesitation degree is assigned to it.

Based on this scale, an Individual Intuitionistic Fuzzy Estimate Vector (IIFEV) is created: $V\tilde{e}^{(l)} = (\tilde{e}_1^{(l)}, \tilde{e}_2^{(l)}, \dots, \tilde{e}_m^{(l)})^T$, where $\tilde{e}_i^{(l)} = (\mu_i^{(l)}, \nu_i^{(l)})$ is the intuitionistic fuzzy estimate of the i th user story rated by the l th expert at a given round. When all IIFEV $V\tilde{e}^{(l)}$ are identical, then we have a unanimous agreement. However, such a situation rarely occurs from the first round with a group having different experiences, skills, and estimation accuracies.

6.4 Aggregation of Individual Intuitionistic Fuzzy Estimation Vectors

The IIFEVs are aggregated at this step to create a single Aggregate Intuitionistic Fuzzy Estimate Vector for the group (AIFEV): $V\tilde{e}^* = (\tilde{e}_1^*, \tilde{e}_2^*, \dots, \tilde{e}_m^*)^T$.

The IFWA operator proposed by [28] is used for this purpose:

$$\begin{aligned} \tilde{e}_i^* &= (\mu_i^*, \nu_i^*) = IFWA_{\lambda} \left(\tilde{e}_i^{(1)}, \tilde{e}_i^{(2)}, \dots, \tilde{e}_i^{(m)} \right) \\ &= \lambda_1 \tilde{e}_i^{(1)} \oplus \lambda_2 \tilde{e}_i^{(2)} \oplus \dots \oplus \lambda_m \tilde{e}_i^{(m)} \\ &= \left(1 - \prod_{l=1}^m (1 - \mu_i^{(l)})^{\lambda_l}, \prod_{l=1}^m (\nu_i^{(l)})^{\lambda_l} \right) \end{aligned} \quad (13)$$

Eventually, when all intuitionistic fuzzy individual estimates $\tilde{e}_i^{(l)} = (\mu_i^{(l)}, \nu_i^{(l)})$ of a story are identical, their aggregation



$\tilde{e}_i^* = (\mu_i^*, v_i^*)$ will be too. The proof is presented in appendix A.

6.5 Consensus Checking

Combining estimates is a convenient way to avoid inconsistency in an expert judgment method, as suggested by Stein and Jørgensen [14]. However, Cohn claimed that the average of individual estimates is not a judicious choice in ASD [50]. Some relevant details about stories, which might be enlightened through discussion, could be lost. For this reason, the aggregate estimate herein is never considered a final estimate, but it is used to check whether the experts' estimates are not divergent before being defuzzified. To check consensus, a similarity distance (also called a consensus index) is defined to calculate the spread between AIFEV and IIFEVs. When all IIFEVs' distances from the AIFEV are below a certain threshold, the consensus is obtained and AIFEV is of acceptable similarity. Based on Eq. (2), the similarity distance between AIFEV and IIFEV is calculated as follows:

$$\begin{aligned} d(V\tilde{e}^*, V\tilde{e}^{(l)}) &= \frac{1}{p} \sum_{i=1}^p d(\tilde{e}_i^*, \tilde{e}_i^{(l)}) \\ &= \frac{1}{2p} \sum_{i=1}^p |\mu_i^* - \mu_i^{(l)}| + |v_i^* - v_i^{(l)}| + |\pi_i^* - \pi_i^{(l)}| \quad (14) \end{aligned}$$

where p is the number of user stories to estimate. Besides, $d(V\tilde{e}^*, V\tilde{e}^{(l)})$ respects the following properties:

1. $0 \leq d(V\tilde{e}^*, V\tilde{e}^{(l)}) \leq 1$;
2. $d(V\tilde{e}^*, V\tilde{e}^{(l)}) = d(V\tilde{e}^{(l)}, V\tilde{e}^*)$;
3. if $d(V\tilde{e}^*, V\tilde{e}^{(l)}) = 0$, then $V\tilde{e}^*$ and $V\tilde{e}^{(l)}$ are completely similar.
4. if $d(V\tilde{e}^*, V\tilde{e}^{(l)}) = 1$, then $V\tilde{e}^*$ and $V\tilde{e}^{(l)}$ are completely dissimilar.

Definition 5 Let $V\tilde{e}^{(l)}$ be the IIFEV of the l th expert and $V\tilde{e}^{(*)}$ be the AIFEV. The consensus is reached when $\forall l \in (1, 2, \dots, m)$, all similarity distances between $V\tilde{e}^{(l)}$ and $V\tilde{e}^{(*)}$ respect the condition:

$$d(V\tilde{e}^*, V\tilde{e}^{(l)}) \leq \tau, \text{ where } 0 \leq \tau \leq 1 \quad (15)$$

6.6 Iterative Algorithm to Reach the Consensus

When conflict arises between estimators, it would often be difficult, even impossible, to reach a consensual estimate. Many causes of this conflict could occur during ASD, such as frustration, stress, pressure, burnout, and belief in superiority. In such circumstances, the consensus should be reached mathematically. An iterative algorithm based on a nonlinear

programming model is performed. The objective is to minimize the similarity distances between AIFEV and IIFEVs in an iterative way. The nonlinear programming model computes the adjustment coefficients required to make AIFEV of an acceptable similarity. Those coefficients could be seen as weights assigned to the stories whose estimates do not assist consensus fulfillment.

In the literature, an iterative algorithm of [63] was proposed with no change in the collective estimate. However, an input coefficient is required to perform the algorithm. In this case, inappropriate values may increase the number of iterations to reach a consensus. Nonetheless, in this study, any additional inputs are required. To do that, let D be a finite set of user stories that do not contribute to the consensus reaching.

$$D = \{u_i | \exists l \in [1, m], \tilde{e}_i^{(l)} \neq \tilde{e}_i^* \wedge d(V\tilde{e}^*, V\tilde{e}^{(l)}) > \tau\} \quad (16)$$

When $\|D\| > 1$, all stories which belong to this set will be iteratively adjusted using adjustment coefficients c_i that minimize the similarity distances as follows:

$$\begin{aligned} \tilde{e}_i^{(l,j+1)} &= (\mu_i^{(l,j+1)}, v_i^{(l,j+1)}) \\ &= (c_i \mu_i^{(l,j)} + (1 - c_i) \mu_i^{*(j)}, c_i v_i^{(l,j)} + (1 - c_i) v_i^{*(j)}) \quad (17) \end{aligned}$$

Indeed, $\tilde{e}_i^{(l,j+1)}$ is the adjusted estimate of the i th user story made by the l th expert at the iteration $j + 1$. It is derived from both his own estimate and the aggregated one of the previous iteration j , using c_i . For this reason, we have to solve a nonlinear programming model that aims to minimize the sum of weighted similarity distances between IIFEVs and AIFEV. Therefore, the following nonlinear programming model is proposed:

$$\min \sum_{u_i \in D} c_i^2 \left(\sum_{l=1}^m d(\tilde{e}_i^*, \tilde{e}_i^{(l)}) \right) \quad (18)$$

$$\text{subject to: } \sum_{u_i \in D} c_i = 1, 0 < c_i < 1.$$

The basic principle of determining those coefficients is that the smaller the similarity distance between individual estimates of a story and the aggregate estimate, the bigger the coefficient should be assigned to this story. Otherwise, the smaller the coefficient should be assigned to it. We can establish the following Lagrangian function:

$$L(c_i, \xi) = \sum_{u_i \in D} c_i^2 \sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)}) - 2\xi \sum_{u_i \in D} c_i - 1 \quad (19)$$



where ξ is a Lagrange multiplier. By differentiating Eq. (19), the coefficients are obtained as follows:

$$c_i = \frac{\frac{1}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})}}{\sum_{u_i \in D} \frac{1}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})}} \quad (20)$$

c_i are strict local minimums. The proof is presented in Appendix B. However, when $\|D\| = 1$, the nonlinear programming model could not be applied through c_i since D contains only one story. Therefore, the iterative adjustment is made differently using experts' weights as follows:

$$\begin{aligned} \tilde{e}_i^{(l,j+1)} &= (\mu_i^{(l,j+1)}, v_i^{(l,j+1)}) \\ &= (\lambda_l \mu_i^{(l,j)} + (1-\lambda_l) \mu_i^{*(j)}, \lambda_l v_i^{(l,j)} + (1-\lambda_l) v_i^{*(j)}) \end{aligned} \quad (21)$$

For each iteration, AIFEV and IIFEVs are adjusted and similarity distances are rechecked. The iterative algorithm stops earlier when the consensus is reached, not up to unanimity. Notice that the iterative algorithm is convergent, and from an iteration n_0 , all adjusted individual estimates will converge to the collective estimate. In this way, we set the following theorem:

Theorem 1 Let $(\mu_i^{(l,j)}, v_i^{(l,j)})$ be the individual intuitionistic fuzzy estimation of the i th user story rated by the l th expert at the j th iteration, for all $j \in \mathbb{N}$ and $l = (1, 2, \dots, m)$, we have:

$\mu_i^{(l,j)}$ and $v_i^{(l,j)}$ are convergent sequences.
 $\exists n_0 \in \mathbb{N}, \lim_{j \rightarrow +\infty} \mu_i^{(l,j)} = \mu_i^{*(n_0)}, \lim_{j \rightarrow +\infty} v_i^{(l,j)} = v_i^{*(n_0)}$

The proof of Theorem 1 is presented in Appendix C.

6.7 Defuzzification of the Collective Estimation into Fibonacci Values

Once the consensus is met on the intuitionistic fuzzy collective estimation either directly or automatically, a defuzzification is required to get a final estimate. In the literature, many score functions have been proposed to defuzzify IFNs, such as [64,65] for interval-valued IFNs and [66] for IFNs. To the most of our knowledge, all proposed researches convert IFNs into crisp real values.

However, as far as we know, the story point is a relative measure that allows comparing the sizes of works with integer values. Besides, the majority of academics and practitioners of agile projects are more familiar with the Fibonacci scale. For this reason, it is better suited to express story points on this scale. Hence, based on the score function of Eq. (11), we propose a formula that converts each intuitionistic fuzzy

aggregate estimate $\tilde{e}_i^* = (\mu_i^*, v_i^*)$ of a story u_i in the consensual collective estimate into a story point SP_i expressed on the Fibonacci scale:

$$SP_i = X + d(X) \times \min(|X - X_1|, |X - X_2|) \quad (22)$$

where $SP_i \in \{0, 1, 2, 3, 5, 8\}$ and $X_1 \leq X \leq X_2$.

$X = S(\tilde{e}_i^*) \times 10$ is the score function of $\tilde{e}_i^*$ multiplied by 10. X_1 is the greatest value of the Fibonacci scale that is inferior or equal to X , and X_2 is the smallest value of the Fibonacci scale that is superior or equal to X . Here, the function $d(X)$ returns a direction sign as follows:

$$\begin{aligned} d(X) : [X_1, X_2] &\longrightarrow \{-1, +1\} \\ d(X) &= \begin{cases} -1 & \text{if } |X - X_1| \leq |X - X_2| \\ +1 & \text{if } |X - X_1| > |X - X_2| \end{cases} \end{aligned} \quad (23)$$

For example, if $X = 4.40$, then according to the Fibonacci scale $X_1 = 3, X_2 = 5, \min(|4.40 - 3|, |4.40 - 5|) = 0.6$ and $d(4.40) = +1$. Therefore, $SP = 5$. In particular, if $X < 1$, then $SP = 0$ or $SP = 1$ depending on Eq. (22). The 0-point story is equivalent to a free lunch [10]. The principle idea of the proposed conversion is assigning the nearest value within the Fibonacci scale to the crisp estimate obtained by the score function. This operation seems to put each user story in the corresponding bucket.

7 Illustrative Example

Assume $D = \{DM_1, DM_2, DM_3\}$ three experts constituting an agile team and responsible for the effort estimation of a collection composed of five user stories $U = \{u_1, u_2, u_3, u_4, u_5\}$. The following are the different steps of the IFEJM:

Step 1: The moderator of the estimate session provides pair-wise comparison matrices using IFAHP and the scale for rating the importance of factors and experts described previously in Table 2. Firstly, the four human factors $F = \{EXP, ASK, PSK, IAE\}$ are compared to each other to create a pair-wise comparison matrix of factors as shown in Table 5. In this illustrative example, experience (EXP) is supposed more important than analysis skills (ASK) and programming skills (PSK), while individual accuracy of estimation (IAE) is very important than any kind of skill.

Tables 6, 7, 8 and 9 represent the pair-wise comparison matrices of experts with respect to the four human factors. The principal information used to carry out those matrices is DM_1 has more experience and estimation accuracy than DM_2 and DM_3 . Regarding analysis skills, DM_1 and DM_2 are almost equally preferred, and both are strongly preferred to DM_3 . Concerning DM_3 , his programming skills



Table 5 Human factors comparison matrix and derived priorities

	EXP	ASK	PSK	IAE	Priority
EXP	(0.5,0.5)	(0.7,0.1)	(0.6,0.1)	(0.3,0.4)	(0.416,0.142)
ASK	(0.1,0.7)	(0.5,0.5)	(0.4,0.5)	(0.1,0.9)	(0.158,0.388)
PSK	(0.1,0.6)	(0.5,0.4)	(0.5,0.5)	(0.1,0.8)	(0.169,0.343)
IAE	(0.4,0.3)	(0.9,0.1)	(0.8,0.1)	(0.5,0.5)	(0.465,0.127)

CI = 0.0527

Table 6 Experts comparison matrix with respect to EXP and derived priorities

	DM_1	DM_2	DM_3	Priority
DM_1	(0.5,0.5)	(0.8,0.1)	(0.9,0.1)	(0.545,0.137)
DM_2	(0.1,0.8)	(0.5,0.5)	(0.6,0.4)	(0.246,0.393)
DM_3	(0.1,0.9)	(0.4,0.6)	(0.5,0.5)	(0.209,0.470)

CI = 0.0888

Table 7 Experts comparison matrix with respect to ASK and derived priorities

	DM_1	DM_2	DM_3	Priority
DM_1	(0.5,0.5)	(0.4,0.5)	(0.7,0.1)	(0.396,0.240)
DM_2	(0.5,0.4)	(0.5,0.5)	(0.9,0.1)	(0.461,0.219)
DM_3	(0.1,0.7)	(0.1,0.9)	(0.5,0.5)	(0.143,0.542)

CI = 0.0986

are better than the other experts. The priority column in these tables has been calculated using Eqs. (6) and (7). Furthermore, all pair-wise comparison matrices are considered consistent since their multiplicative consistency indexes (calculated using Eq. (5)) are less than 0.1. The composite weights of experts are determined using Eq. (12) as follows: $\lambda = \{\lambda_1 = 0.3749, \lambda_2 = 0.3244, \lambda_3 = 0.3007\}$.

Step 2: The agile team sets the max number of rounds $maxR = 3$ and the consensus threshold $\tau = 0.05$.

Step 3: The first round begins ($r = 1$), and each expert is invited to give his IIFEV $V\tilde{e}^{(l)}$ of rating the five uses stories

Table 8 Experts comparison matrix with respect to PSK and derived priorities

	DM_1	DM_2	DM_3	Priority
DM_1	(0.5,0.5)	(0.4,0.5)	(0.2,0.8)	(0.250,0.424)
DM_2	(0.5,0.4)	(0.5,0.5)	(0.2,0.7)	(0.271,0.377)
DM_3	(0.8,0.2)	(0.7,0.2)	(0.5,0.5)	(0.479,0.200)

CI = 0.0746

Table 9 Experts comparison matrix with respect to IAE and derived priorities

	DM_1	DM_2	DM_3	Priority
DM_1	(0.5,0.5)	(0.6,0.1)	(0.9,0.1)	(0.548,0.154)
DM_2	(0.1,0.6)	(0.5,0.5)	(0.4,0.3)	(0.241,0.378)
DM_3	(0.1,0.9)	(0.3,0.4)	(0.5,0.5)	(0.212,0.469)

CI = 0.0414



using Table 4:

$$V\tilde{e}^{(1)} = \begin{pmatrix} HE \\ ME \\ LE \\ ME \\ ELE \end{pmatrix} V\tilde{e}^{(2)} = \begin{pmatrix} LE \\ VLE \\ ME \\ HE \\ ME \end{pmatrix} V\tilde{e}^{(3)} = \begin{pmatrix} ELE \\ LE \\ HE \\ ELE \\ ME \end{pmatrix}$$

Steps 4 and 5: All experts' estimates are revealed; use the IFWA operator of Eq. (13) to calculate the AIFEV $V\tilde{e}^*$; and check the similarity distances between the latter and each IIFEV using Eq. (14).

$$V\tilde{e}^* = \begin{pmatrix} (0.4381, 0.3553) \\ (0.3047, 0.5796) \\ (0.4851, 0.3163) \\ (0.4760, 0.3256) \\ (0.3385, 0.5421) \end{pmatrix}$$

$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.2415$, $d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.1933$, and $d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.3096$. We can see that all similarity distances are greater than $\tau = 0.05$ and hence go to the next round $r = r + 1$ as long as $r \leq 3$. After discussion, the experts revise their initial opinions and make a second round of estimation:

$$V\tilde{e}^{(1)} = \begin{pmatrix} HE \\ ME \\ LE \\ ME \\ ELE \end{pmatrix} V\tilde{e}^{(2)} = \begin{pmatrix} ME \\ LE \\ ME \\ LE \\ ME \end{pmatrix} V\tilde{e}^{(3)} = \begin{pmatrix} LE \\ ME \\ LE \\ ELE \\ ELE \end{pmatrix}$$

The AIFEV $V\tilde{e}^*$ and the similarity distances are recalculated:

$$V\tilde{e}^* = \begin{pmatrix} (0.5190, 0.2753) \\ (0.3918, 0.4682) \\ (0.3962, 0.4629) \\ (0.4381, 0.3553) \\ (0.2329, 0.6918) \end{pmatrix}$$

$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.1564$, $d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.1912$, and $d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.2766$. Despite decreasing the similarity distances compared to the first round, they are still all greater than the consensus threshold. For that, another round is required ($r = r + 1$). The following are the estimates for the third round:

$$V\tilde{e}^{(1)} = \begin{pmatrix} ME \\ LE \\ ME \\ HE \\ VLE \end{pmatrix} V\tilde{e}^{(2)} = \begin{pmatrix} ME \\ ME \\ HE \\ LE \end{pmatrix} V\tilde{e}^{(3)} = \begin{pmatrix} ME \\ ELE \\ HE \\ LE \\ VLE \end{pmatrix}$$

The corresponding AIFEV $V\tilde{e}^*$ and similarity distances of this round:

$$V\tilde{e}^* = \begin{pmatrix} (0.4500, 0.4000) \\ (0.2836, 0.6124) \\ (0.6235, 0.1682) \\ (0.6048, 0.1756) \\ (0.1838, 0.7479) \end{pmatrix}$$

$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.0833$, $d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.0964$, and $d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.1781$. The collective estimate is of unacceptable similarity, although the similarity distances are closer to 0.05. In the meantime, the number of rounds attains the $\max R$ (i.e., $r = 3$), and the consensus is not met. Hence, the iterative algorithm should be performed.

Step 6: Determine the user stories which do not contribute to the consensus reaching. We can see clearly through the individual estimations vectors of the last round that the user stories $D = \{u_2, u_3, u_4, u_5\}$ are the target user stories. Since $\|D\| > 1$, apply the iterative algorithm based on the nonlinear programming model to find the adjustment coefficients of these stories. Using Eq. (20), we have: $c_2 = 0.1705$, $c_3 = 0.2381$, $c_4 = 0.1379$, and $c_5 = 0.4535$. Afterward, based on Eq. (17), use these coefficients to adjust iteratively estimates:

$$V\tilde{e}^{(1)} = \begin{pmatrix} (0.4500, 0.4000) \\ (0.2778, 0.6188) \\ (0.5822, 0.2234) \\ (0.6179, 0.1652) \\ (0.1685, 0.7715) \end{pmatrix} V\tilde{e}^{(2)} = \begin{pmatrix} (0.4500, 0.4000) \\ (0.3119, 0.5762) \\ (0.6417, 0.1519) \\ (0.6179, 0.1652) \\ (0.2138, 0.7035) \end{pmatrix}$$

$$V\tilde{e}^{(3)} = \begin{pmatrix} (0.4500, 0.4000) \\ (0.2523, 0.6614) \\ (0.6417, 0.1519) \\ (0.5559, 0.2410) \\ (0.1685, 0.7715) \end{pmatrix} V\tilde{e}^* = \begin{pmatrix} (0.4500, 0.4000) \\ (0.2816, 0.6169) \\ (0.6204, 0.1755) \\ (0.6003, 0.1850) \\ (0.1835, 0.7488) \end{pmatrix}$$

$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.0188$, $d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.0259$, and $d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.0294$. The iterative algorithm stops after only performing one iteration since all similarity distances are less than 0.05. Consequently, the collective estimate is of acceptable similarity. Moreover, the experiment shows that unanimity could be reached after 23 iterations.

Step 7: Defuzzify the consensual collective estimate vector by calculating the corresponding story points on the Fibonacci scale. Using Eq. (22), the resulting crisp consensual collective estimate vector is as follows:

$$Ve^* = \begin{pmatrix} 5 \\ 3 \\ 8 \\ 8 \\ 2 \end{pmatrix}$$

8 Experiment and Results

The dataset used in our experiment was previously published in [67], and it concerns an agile project with three sprints. The user stories in the two first sprints were estimated by students, while those of the third one were estimated by professionals. The authors of [67] used a Bayesian network (BN) model and the conventional Preprocessing Poker (PP) to provide estimates.

A dataset preprocessing is required before performing experiments. The preprocessing consists of (i) rounding down the estimate of tall stories, (ii) converting estimates to IFNs, and (iii) gathering stories by lots and selecting the



highest estimate as a group estimate. Notice that the post-estimates have been taken as actual efforts because of the experience gained by estimators after developing the user stories. The consensus threshold τ is set to 0.05.

Usually, the mean magnitude of relative error MMRE and the prediction percentage PRED are used to compare models' accuracies. However, they have often been criticized in the literature for their lack of concern for underestimates and overestimates in calculating model accuracy [68–72]. Instead, the Balanced of Relative Error (BRE) is used. The BRE establishes a balance between underestimates and overestimates. There is even a BRE-based measure, so-called BREbias. This measure indicates whether an assessment is toward overestimation (negative BREbias) or underestimation (positive BREbias). Furthermore, BRE and BREbias have been used in many research studies focused on effort estimation in ASD [68,72,73]. The two measures are calculated as follows:

$$BRE = \frac{|Actual_Effort - Estimated_Effort|}{\min(Actual_Effort, Estimated_Effort)}$$

$$BREbias = \frac{Actual_Effort - Estimated_Effort}{\min(Actual_Effort, Estimated_Effort)}$$

It was also claimed that absolute residual-based measures are better for accuracy comparison between prediction models than MMRE and PRED [70,71]. Herein, we use further the following absolute residual-based measures: the mean of absolute residuals (Mean Ab.Res) to assess accuracy, the median of absolute residuals (Med Ab. Res), and the standard deviation of absolute residuals (SD Ab.Res) to evaluate deviation from actual estimates. The absolute residual measure is calculated by:

$$Abs.Res = |Actual_Effort - Estimated_Effort|$$

Moreover, given that we have small sizes of samples and data are not normally distributed, the nonparametric Wilcoxon signed-rank test is chosen to prove the significance of our outcomes.

8.1 Experiment Based on Student Estimates

The first experiment was conducted on the 1st sprint and the 2nd sprint of the agile project, having nine and five stories respectively. The estimation was made by five students from the computer engineering undergraduate program. In the first sprint, the intuitionistic collective estimate and similarity dis-

tances are obtained as follows:

$$V\tilde{e}^* = \begin{pmatrix} (0.3674, 0.4595) \\ (0.2115, 0.7063) \\ (0.1933, 0.7404) \\ (0.1203, 0.8586) \\ (0.2024, 0.7231) \\ (0.6613, 0.1320) \\ (0.1615, 0.7858) \\ (0.2772, 0.6149) \\ (0.2504, 0.6562) \end{pmatrix}$$

$$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.0695, d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.1486, \\ d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.1417, d(V\tilde{e}^*, V\tilde{e}^{(4)}) = 0.1276, \text{ and } \\ d(V\tilde{e}^*, V\tilde{e}^{(5)}) = 0.1376.$$

In the second sprint, the intuitionistic collective estimate and similarity distances are obtained as follows:

$$V\tilde{e}^* = \begin{pmatrix} (0.1567, 0.7938) \\ (0.5245, 0.2854) \\ (0.1591, 0.7914) \\ (0.1741, 0.7656) \\ (0.1796, 0.7584) \end{pmatrix}$$

$$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.0774, d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.0746, \\ d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.0546, d(V\tilde{e}^*, V\tilde{e}^{(4)}) = 0.1218, \text{ and } \\ d(V\tilde{e}^*, V\tilde{e}^{(5)}) = 0.2172.$$

From similarity distances, we can observe that consensus could not be reached for students in both sprints. For this reason, the iterative algorithm has been applied. After Adjustment and defuzzification, the crisp consensual collective estimate for Sprint 1 is

$$Ve^* = \begin{pmatrix} 5 \\ 2 \\ 2 \\ 1 \\ 2 \\ 2 \\ 8 \\ 2 \\ 3 \\ 3 \end{pmatrix}$$

With the adjustment coefficients: $c_1 = 0.0597$, $c_2 = 0.2453$, $c_3 = 0.1481$, $c_4 = 0.3621$, $c_5 = 0.1848$, $c_6 = 0.2549$, $c_7 = 0.3680$, $c_8 = 0.2132$, and $c_9 = 0.1640$.

On the other hand, the crisp consensual collective estimate for Sprint 2 is

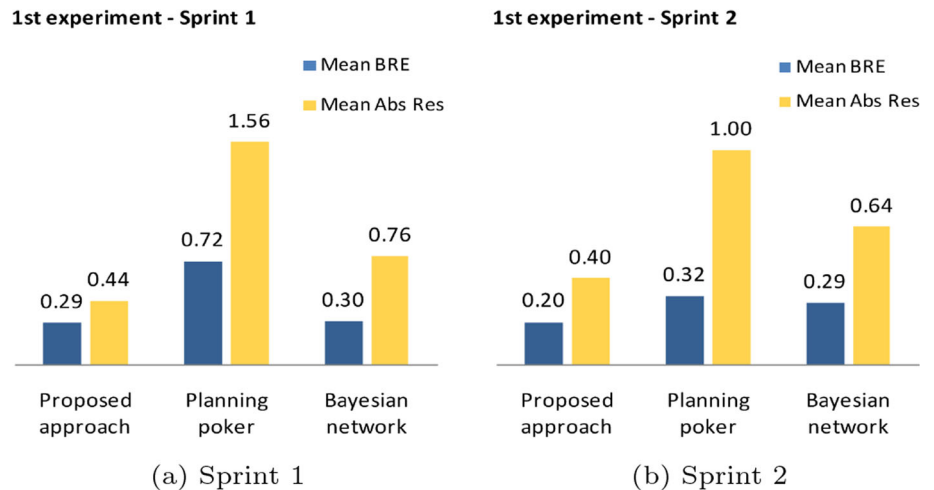
$$Ve^* = \begin{pmatrix} 2 \\ 5 \\ 2 \\ 2 \\ 2 \end{pmatrix}$$

With the adjustment coefficients: $c_1 = 0.2969$, $c_2 = 0.1245$, $c_3 = 0.2127$, $c_4 = 0.1845$, and $c_5 = 0.1815$.

Regarding models' comparison, Table 10 shows that IFEJM has attained a mean BRE value of 0.29, a mean Abs. Res of 0.44 in sprint 1, while a mean BRE value of 0.20 and a mean Abs.Res of 0.40 in sprint 2. In both sprints, the

Table 10 Estimation accuracy of the studied models by sprint in the first experiment

Sprint	No. of user stories	Model	BRE		Abs.Res		
			Mean	Median	Mean	Median	SD
Sprint 1	9	Proposed approach	0.29	0.00	0.44	0.00	0.87
		PP technique	0.72	−0.50	1.56	1.00	2.24
		BN model	0.30	−0.10	0.76	0.40	1.19
Sprint 2	5	Proposed approach	0.20	0.00	0.40	0.00	0.71
		PP technique	0.32	−0.50	1.00	1.00	1.66
		BN model	0.29	0.25	0.64	0.60	0.77

Fig. 3 Comparison of mean BRE and mean Abs.Res of the studied models in the two sprints of the first experiment

mean BRE and mean Abs.Res of the proposal are less than those of PP and BN. This confirms that it has better estimation accuracy than the other studied models. Moreover, the proposal presents a null bias toward overestimating and underestimating in both sprints (median BREbias = 0.00 and median Abs.Res = 0.00). Meanwhile, PP shows a high overestimation bias (median BREbias = −0.50) overall stories of the two sprints. BN has a low overestimation bias in sprint 1 (median BREbias = −0.10) and moderate underestimation bias in sprint 2 (median BREbias = 0.25).

The SD Ab.Res indicates that the poor BREbias performances obtained by PP and BN are due to the impact of outliers. Overall, for both sprints estimated by students, the proposal outcomes are the best among all the three studied models, as shown in Fig. 3.

Furthermore, the BRE and Abs.Res of the studied estimate models are not normally distributed. For instance, in sprint 1 the BRE histograms of the studied models (see Fig. 4) are not symmetrical about the mean and did not peak in the middle. As well, the boxplots of Abs.Res (see Fig. 5) in both sprints are not symmetrical and contain outliers. The graphical normality testing is also approved by the Shapiro–Wilk test using R. Accordingly, the nonparametric Wilcoxon signed-rank test is applied with the significance level $\alpha = 0.05$ to statically test the significance of our outcomes.

However, since the size of the studied dataset is small, the results should be carefully interpreted. The Wilcoxon signed-rank test applied to Abs.Res and BREbias (see Tables 11 and 12) shows strong evidence that the proposed approach is quite accurate than PP in the 1st sprint ($p = 0.0185$ at $\alpha = 0.05$) with a significant difference in BREbias ($p = 0.0132$ at $\alpha = 0.05$). In the meantime, there is no significant difference with the BN for both sprints. In the 2nd sprint, it is still a significant difference between the proposal's BREbias and the PP's BREbias ($p = 0.0477$ at $\alpha = 0.05$). This confirms that the proposal maintains a better bias for both sprints compared to the other models. Despite the proposed approach providing better outcomes than those of the other models in sprint 2, they are not good enough to be significant. This fact could be explained by the enhancement of students' estimates in sprint 2, and the growth of group agreement at this sprint. Another reason is tied to the sample size which is not enough to draw statistical conclusions.

8.2 Experiment Based on Experts Estimates

The second experiment was carried out on the third sprint with six user stories estimated by four experts. Since experts have not provided post-estimates in this experiment, the proposed approach is compared to the PP technique according



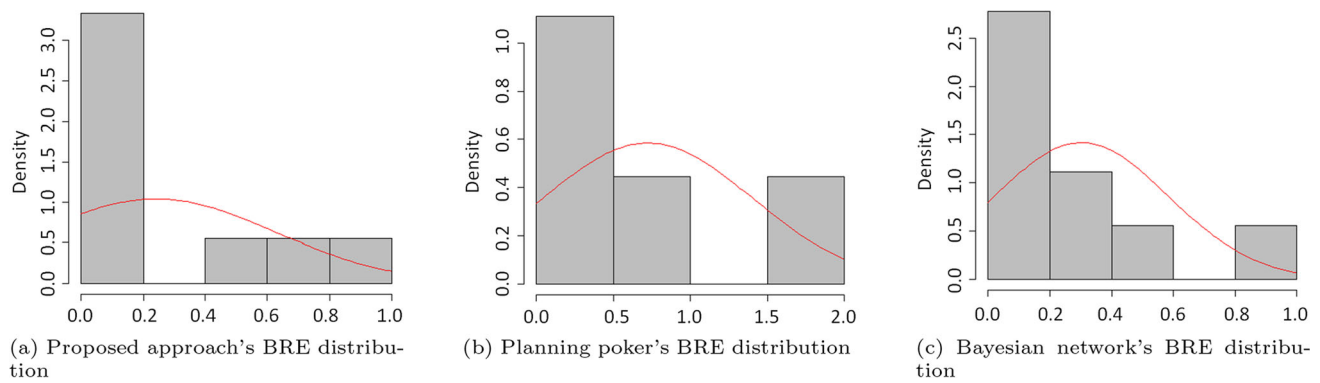


Fig. 4 Normality testing by BRE histograms in sprint 1

Fig. 5 Normality testing by Abs.Res boxplots in both sprints

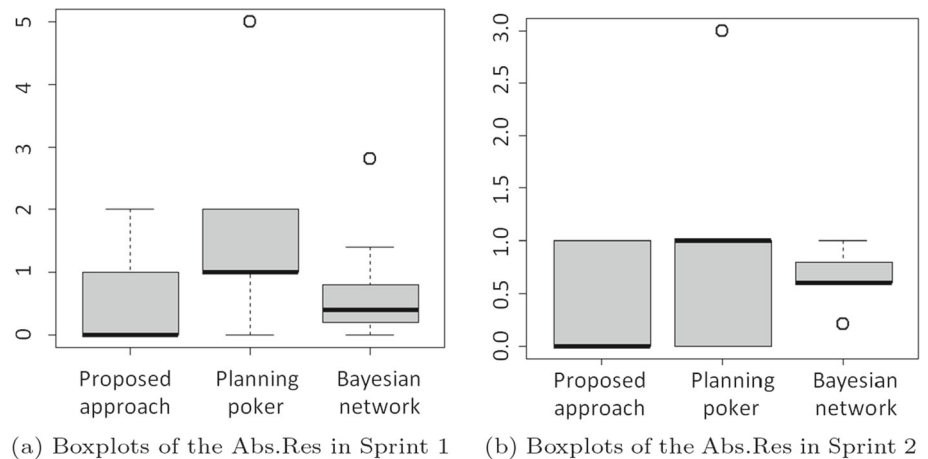


Table 11 The Wilcoxon signed-rank test of the absolute residuals for sprints 1 and 2

Hypothesis	Sprint 1		Sprint 2	
	Z-score	p value	Z-score	p value
H0: Proposed approach's mean Abs.Res $\geq$ PP's mean Abs.Res	-2.0847	0.0185	-0.5656	0.2858
H0: Proposed approach's mean Abs.Res $\geq$ BN's mean Abs.Res	-0.4962	0.3098	-0.9128	0.1806

to the estimates of the trained BN. By applying the IFEJM, the intuitionistic collective estimate and similarity distances are obtained as follows:

$$V\tilde{e}^* = \begin{pmatrix} (0.7000, 0.1000) \\ (0.4500, 0.4000) \\ (0.5273, 0.2828) \\ (0.2262, 0.6846) \\ (0.1500, 0.8000) \\ (0.1500, 0.8000) \end{pmatrix}$$

$$d(V\tilde{e}^*, V\tilde{e}^{(1)}) = 0.0388, d(V\tilde{e}^*, V\tilde{e}^{(2)}) = 0.0253, d(V\tilde{e}^*, V\tilde{e}^{(3)}) = 0.0362, \text{ and } d(V\tilde{e}^*, V\tilde{e}^{(4)}) = 0.0253.$$

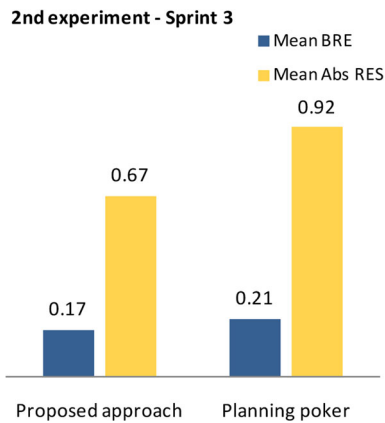
In contrast to the first experiment, the consensus is reached without applying the iterative algorithm since the experts' estimates are very close. This fact is confirmed by similarity distances that are all below the threshold. Thereby, the collective estimate could be defuzzified, and the crisp consensual

Table 12 The Wilcoxon signed-rank test of the BREbias for sprints 1 and 2

Hypothesis	Sprint 1		Sprint 2	
	Z-score	p value	Z-score	p value
H0: Proposed approach's BREbias = PP's BREbias	2.4779	0.0132	1.9798	0.0477
H0: Proposed approach's BREbias = BN's BREbias	-0.2103	0.8334	0.1825	0.8551

Table 13 Estimation accuracy of the studied models in the second experiment

Sprint	No. of user stories	Model	BRE		Abs.Res		
			Mean	Median	Mean	Median	SD
Sprint 3	6	Proposed approach	0.17	0.00	0.67	0.38	1.15
		PP technique	0.21	0.00	0.92	0.43	1.49

**Fig. 6** Comparison of mean BRE and mean Abs.Res between proposed approach and planning poker

one for this third sprint is

$$Ve^* = \begin{pmatrix} 8 \\ 5 \\ 5 \\ 3 \\ 2 \\ 2 \end{pmatrix}$$

Regarding models' comparison, Table 13 shows that IFEJM has attained the mean BRE value of 0.17, the mean Abs.Res value of 0.67, with a null bias toward overestimation and underestimation (median BREbias = 0.00). Those values are better than those of the PP as shown in Fig. 6. Given that all experts' estimates of the third sprint are close to each other, there is no significant difference approved by the nonparametric Wilcoxon signed-rank test.

8.3 Discussion

First of all, regarding data preprocessing, the estimate conversion into linguistic terms representing IFNs is an effective tool to deal with problems that involve imprecise and vague data, notably effort estimation. By doing so, estimators could rate the stories with a limited universe of discourse instead of the Fibonacci sequence numbers. Commonly, humans feel more comfortable with linguistic terms to make judgments [24].

Beyond the communication aspect that has been marginalized due to the absence of many estimation rounds in the

dataset, the consensus threshold is set as small as possible. (τ is set to 0.05.) For that, the final estimate is not the average of the individual ones [50]. The experiment has also confirmed that the team's highest estimate is not a convenient way to combine estimates in ASD. Indeed, the studied models PP and BN showed a considerable overestimation bias, especially in the first experiment. This fact confirms that a consensus-based combination, although sometimes obtained mathematically, is the ideal choice for agile projects.

Concerning the human factors, unfortunately, the dataset did not provide any information about the skills and experiences of students and experts, which prevents us from proving the impact of those factors. However, we have integrated individual estimation accuracy as a human factor to make preferences between estimators, especially for students. Preferences between experts could not be made since there is only one sprint in the second experiment. In this way, students in sprint 1 have an initial common weight ($\lambda = 0.2$), while the students' estimates in sprint 2 have been weighted through ($\lambda_1 = 0.14$, $\lambda_2 = 0.19$, $\lambda_3 = 0.22$, $\lambda_4 = 0.25$, and $\lambda_5 = 0.19$) deduced from their prior accuracy of estimation in the previous sprint. The BRE and absolute residuals measures show that the accuracy of our IFEJM is increased in sprint 2, which is obvious because accurate students have been privileged. This fact justifies the usefulness of integrating individual estimation accuracy as a human factor in our proposal.

On the other hand, the two experiments confirmed that our approach supports both behavioral and mathematical consensus. Indeed, in the first experiment, we had to find the mathematical one through the iterative algorithm for both sprints, while the behavioral was directly reached in the second experiment. Only one iteration has been performed in the first experiment to adjust estimates and get a consensus. This highlights the performance of our approach, which aims to quickly get a consensual estimation instead of multiplying rounds to reach a common one by unanimity. Besides, estimating a set of user stories all at once, without exceeding a certain number, (we recommend 5 stories), increases analogy in the assessment and gain of time to reach consensus.

Overall, being conscious of the incompleteness of the case studies used here, the initial results are still however promising. Indeed, it is concluded in the first experiment that the proposal can provide accurate estimation better than PP, especially with inexperienced estimators or in the first sprints



when disagreement is great and estimates are divergent. Furthermore, even if there is no significant evidence that the proposal is better than BN, it is still closer to the students' post-estimates. Besides, despite experts' estimates are convergent in the second experiment, the proposal maintains a null bias toward the overestimation or the underestimation of the user stories.

9 Conclusion and Perspectives

To deal with the consensus problem in the actual solution, it is much faster and easier to get agreement among experts to set a consensus threshold rather than multiplying rounds to reach a common estimate for each user story. Likewise, an upper limit of rounds is set to allow directly reaching behavioral consensus without exceeding this limit. However, when it is impossible to achieve it, a mathematical consensus could be reached through an iterative algorithm based on a nonlinear programming model.

The computation of the consensual collective estimate also integrates the priority weights of experts according to a set of human factors. Thanks to IFAHP, these weights are derived. This study has been focused on the estimator himself because, in our view, an experienced and skilled estimator could predict accurate effort estimation. Moreover, the moderator could define additional sub-factors to make more granular preferences between experts. It is important to note that pair-wise comparison matrices should not all be reproduced again for each estimate session, especially when there is no change in the team. Only the priorities of experts according to their estimation accuracies should be actualized.

The final estimation is obtained through a defuzzification on the Fibonacci scale. A score function that introduces a novel concept so-called voting matrix of an IFN has been used to this end. The study also involves intuitionistic fuzzy logic throughout the estimating process. This choice has been argued by the flexibility of this logic to deal with incertitude and vagueness in expert judgments.

In terms of experiments, the initial results revealed that the proposal is more accurate than the other studied models, especially with inexperienced estimators or in the first sprints of ASD projects. In the rest of the project sprints, the proposal maintains a null bias toward overestimation and underestimation.

As perspectives, the proposal could be applied to other expert judgment problems related to project management in software engineering. For instance, risk assessment, budgetary estimate, assessment of restrictions and technical constraints, etc. For this reason, new factors and rating scales should be defined. In the forthcoming works, we plan to provide more applications and empirical studies of the IFEJM.

A new tool that supports the proposal should, of course, be included in our plans.

Appendix A: All Individual Estimates Are Identical

Proof Replacing all $\mu_i^{(l)}$ by $\mu_i^{(1)}$ and all $v_i^{(l)}$ by $v_i^{(1)}$ in Eq. (13), we have:

$$\begin{aligned}(\mu_i^*, v_i^*) &= IFWA_\lambda \left(\tilde{e}_i^{(1)}, \tilde{e}_i^{(2)}, \dots, \tilde{e}_i^{(m)} \right) \\ &= \left(1 - (1 - \mu_i^{(1)})^{\sum_{l=1}^m \lambda_l}, (v_i^{(1)})^{\sum_{l=1}^m \lambda_l} \right)\end{aligned}$$

Since $\sum_{l=1}^m \lambda_l = 1$, consequently

$$(\mu_i^*, v_i^*) = (\mu_i^{(1)}, v_i^{(1)}) = (\mu_i^{(2)}, v_i^{(2)}) = \dots = (\mu_i^{(m)}, v_i^{(m)})$$

□

Appendix B: Proof of Equation (20)

Proof By differentiating the Lagrangian function in Eq. 19, we can get the derivatives:

$$\begin{cases} \frac{\partial L(c_i, \xi)}{\partial c_i} = 2c_i \sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)}) - 2\xi \\ \frac{\partial L(c_i, \xi)}{\partial \xi} = \sum_{u_i \in D} c_i - 1 \end{cases}$$

According to the Lagrange multiplier method [74], the stationary points of $L(c_i, \xi)$ could be found by setting the above derivatives to zero: $\frac{\partial L(c_i, \xi)}{\partial c_i} = \frac{\partial L(c_i, \xi)}{\partial \xi} = 0$. We obtain afterward the following equations:

$$c_i \sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)}) - \xi = 0 \quad (\text{B1})$$

$$\sum_{u_i \in D} c_i - 1 = 0 \quad (\text{B2})$$

Using Eq. (B1), we have:

$$c_i = \frac{\xi}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})}$$

By replacing c_i in Eq. (B2), we have:

$$\sum_{u_i \in D} \frac{\xi}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})} = 1$$



so that

$$\xi = \frac{1}{\sum_{u_i \in D} \frac{1}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})}}$$

Therefore, the coefficients of adjustment are obtained as follows:

$$c_i = \frac{\frac{1}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})}}{\sum_{u_i \in D} \frac{1}{\sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})}} \quad (\text{B3})$$

On the other hand, the Hessian matrix of Lagrangian function:

$$H = \frac{\partial^2 L(c_i)}{\partial c_i \partial c_j}$$

where

$$\frac{\partial^2 L(c_i)}{\partial c_i \partial c_j} = 0 \text{ for } i \neq j$$

and

$$\frac{\partial^2 L(c_i)}{\partial c_i \partial c_j} = 2 \sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})$$

The eigenvalues of H are $2 \sum_{l=1}^m d_l(\tilde{e}_i^*, \tilde{e}_i^{(l)})$ greater than 0. According to [74], H is positive definite. Thereby, c_i is a strict local minimum. $\square$

Appendix C: Proof of Theorem 1

Proof Using Eq. (17), we consider $\mu_i^{(l,j)}$ as a sequence:

$$\mu_i^{(l,j+1)} = c_i \mu_i^{(l,j)} + (1 - c_i) \mu_i^{*(j)} \quad (\text{C4})$$

Based on the monotonic sequence theorem, every bounded, monotonic sequence converges. First, we demonstrate that $\mu_i^{(l,j)}$ is a bounded sequence, i.e., $0 \leq \mu_i^{(l,j)} \leq 1$ for all $j \in \mathbb{N}$. By definition $\mu_i^{(l,1)}$ is a membership function, and hence, $0 \leq \mu_i^{(l,1)} \leq 1$. Using the principle of mathematical induction, we suppose that $0 \leq \mu_i^{(l,j)} \leq 1$ is true and we will prove that $0 \leq \mu_i^{(l,j+1)} \leq 1$ is also true. Indeed, by

substituting the term j by $j-1$ in Eq. (C4), then we have:

$$\mu_i^{(l,j+1)} = c_i \left(c_i \mu_i^{(l,j-1)} + (1 - c_i) \mu_i^{*(j-1)} \right) + (1 - c_i) \mu_i^{*(j)}$$

so that

$$\mu_i^{(l,j+1)} = c_i^2 \mu_i^{(l,j-1)} + c_i(1 - c_i) \mu_i^{*(j-1)} + (1 - c_i) \mu_i^{*(j)}$$

Next, by substituting the term $j-1$ by $j-2$, we have:

$$\begin{aligned} \mu_i^{(l,j+1)} &= c_i^2 \left(c_i \mu_i^{(l,j-2)} + (1 - c_i) \mu_i^{*(j-2)} \right) \\ &\quad + c_i(1 - c_i) \mu_i^{*(j-1)} + (1 - c_i) \mu_i^{*(j)} \end{aligned}$$

So on, we continue substituting until the first term. We obtain afterward:

$$\begin{aligned} \mu_i^{(l,j+1)} &= c_i^j \mu_i^{(l,1)} + c_i^{j-1} (1 - c_i) \mu_i^{*(1)} + c_i^{j-2} (1 - c_i) \mu_i^{*(2)} \\ &\quad + \dots + c_i(1 - c_i) \mu_i^{*(j-1)} + (1 - c_i) \mu_i^{*(j)} \end{aligned}$$

Accordingly, we write:

$$\mu_i^{(l,j+1)} = c_i^j \mu_i^{(l,1)} + (1 - c_i) \sum_{k=0}^{j-1} c_i^k \mu_i^{*(j-k)} \quad (\text{C5})$$

Obviously, $\mu_i^{(l,j+1)} \geq 0$ since $0 \leq \mu_i^{(l,1)} \leq 1$, $0 < c_i < 1$, and $0 \leq \mu_i^{*(j)} \leq 1$ for all $j \in \mathbb{N}$.

Furthermore, from Eq. (C5) we have:

$$\mu_i^{(l,j+1)} \leq c_i^j \mu_i^{(l,1)} + (1 - c_i) \sum_{k=0}^{j-1} c_i^k \text{ and } \sum_{k=0}^{j-1} c_i^k = \frac{(1 - c_i^j)}{(1 - c_i)}.$$

This implies $\mu_i^{(l,j+1)} \leq c_i^j \mu_i^{(l,1)} + (1 - c_i) \frac{(1 - c_i^j)}{(1 - c_i)}$ and

$\mu_i^{(l,j+1)} \leq c_i^j \mu_i^{(l,1)} + 1 - c_i^j$, i.e., $\mu_i^{(l,j+1)} \leq c_i^j (\mu_i^{(l,1)} - 1) + 1$, since $0 \leq c_i^j \leq 1$ and $\mu_i^{(l,1)} \leq 1 \Rightarrow c_i^j (\mu_i^{(l,1)} - 1) \leq 0$. Hence, $0 \leq \mu_i^{(l,j+1)} \leq 1$.

The first property of Theorem 1 is thus verified. $\mu_i^{(l,j)}$ is a bounded sequence.

Next, we demonstrate that $\mu_i^{(l,j)}$ is a monotonic sequence.

We can see that $\mu_i^{(l,j)}$ is a recursive sequence that could be denoted as:

$$\mu_i^{(l,j+1)} = f(\mu_i^{(l,j)}), \text{ where } f : [0, 1] \longrightarrow [0, 1]$$

and

$$f(x) = c_i x + (1 - c_i) \left(1 - (1 - x)^{\lambda_l} \prod_{l'=1, l' \neq l}^m (1 - \mu_i^{(l',j)})^{\lambda_{l'}} \right) \quad (\text{C6})$$



To simplify the recursion function, we set

$p = \prod_{l'=1, l' \neq l}^m (1 - \mu_i^{(l', j)})^{\lambda_{l'}}$ where $p \in [0, 1]$. Then, the function $f(x)$ in Eq. (C6) could be written as:

$$f(x) = c_i x - (1 - c_i)p(1 - x)^{\lambda_l} + (1 - c_i) \quad (C7)$$

Next, we derive $f(x)$

$$\frac{\partial f(x)}{\partial x} = c_i + \lambda_l(1 - c_i)p \quad (C8)$$

Since $c_i > 0$ and $\lambda_l, p \in [0, 1]$, we have: $\frac{\partial f(x)}{\partial x} > 0$ for every $x \in [0, 1]$. We deduce that $f(x)$ is a strictly increasing function, and consequently, $\mu_i^{(l, j)}$ is a monotonic (increasing or decreasing) sequence. Its monotonicity depends on the sign of $\mu_i^{(l, 2)} - \mu_i^{(l, 1)}$. If $\mu_i^{(l, 2)} - \mu_i^{(l, 1)} > 0$, then $\mu_i^{(l, j)}$ is a strictly increasing sequence. Otherwise, $\mu_i^{(l, j)}$ is a strictly decreasing sequence. From above, we proved that $\mu_i^{(l, j)}$ is a bounded monotonic sequence, and therefore, it converges. Now, it remains only to find its limit.

Based on the theorem of the fixed point: A fixed point l of a continuous recursion function $f(x)$, which satisfies $f(l) = l$ and $|\frac{\partial f(x)}{\partial x}| < 1$, is unique and limit of the recursive sequence expressed by $f(x)$. By using Eq. (C7), we have:

$$f(\mu_i^{*(n_0)}) = c_i \mu_i^{*(n_0)} - (1 - c_i)p(1 - \mu_i^{*(n_0)})^{\lambda_l} + (1 - c_i)$$

We have also: $p(1 - \mu_i^{*(n_0)})^{\lambda_l} = \prod_{l=1}^m (1 - \mu_i^{(l, n_0)})^{\lambda_l}$. Thereby

$$\begin{aligned} f(\mu_i^{*(n_0)}) &= c_i \mu_i^{*(n_0)} - (1 - c_i) \prod_{l=1}^m (1 - \mu_i^{(l, n_0)})^{\lambda_l} + (1 - c_i) \\ &= c_i \mu_i^{*(n_0)} + (1 - c_i) \left(1 - \prod_{l=1}^m (1 - \mu_i^{(l, n_0)})^{\lambda_l}\right) \\ &= c_i \mu_i^{*(n_0)} + (1 - c_i) \mu_i^{*(n_0)} \end{aligned}$$

We find that $f(\mu_i^{*(n_0)}) = \mu_i^{*(n_0)}$. This means that $\mu_i^{*(n_0)}$ is a fixed point of $f(x)$. From Eq. (C8), by applying the triangular inequality, we have:

$$\left| \frac{\partial f(x)}{\partial x} \right| \leq |c_i| + \lambda_l p |1 - c_i| < |c_i| + |1 - c_i|$$

Since the only case that $\lambda_l p |1 - c_i| = |1 - c_i|$ is when $p = 1$ and $\lambda_l = 1$, which is absurd because if we admit that $\lambda_l = 1$, then $p=0$. This means that $|\frac{\partial f(x)}{\partial x}| \leq 1$. Now, recall that since $f(x)$ is differentiable on the interval $[0, 1]$, then it is certainly continuous on this interval. From above, we proved that $\mu_i^{*(n_0)}$ is a unique fixed point of the continue

function $f(x)$ and furthermore the limit of its corresponding recursive sequence $\mu_i^{(l, j)}$. Hence, we can write:

$$\lim_{j \rightarrow +\infty} \mu_i^{(l, j)} = \mu_i^{*(n_0)}$$

Likewise, we can demonstrate $v_i^{(l, j)}$ is a convergent sequence with

$$\lim_{j \rightarrow +\infty} v_i^{(l, j)} = v_i^{*(n_0)}$$

This completes the proof of Theorem 1. Obviously, Theorem 1 is still valid despite the set D contains only one user story, simply we have to replace c_i by λ_l . $\square$

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